

Himani Arora, PhD

+1 510-424-3163
✉ himani.arora.06@gmail.com
in himani-arora-
himani-arora-ha.github.io
Google Scholar

Senior Process Engineer, Epitaxy, Applied Materials

Professional Experience

Since 2022 **Sr. Process Engineer**, Applied Materials (300mm Semiconductor Epitaxy)

- Over 5 years of experience in semiconductor processing and device fabrication, including field-effect transistors, photodetectors, and solar cells.
- Develop and optimize epitaxial growth processes for FEOL applications, aligned with device performance requirements.
- Conduct material and structural analysis using XRD, RSM, ellipsometry, four-probe resistivity, SIMS, SEM, and TEM to support process improvement.
- Lead process development and sustainment in R&D and customer-facing projects, including on-site tool installations.
- Apply statistical process control (SPC) and data analysis to improve process stability and yield.
- Provide technical training on cleanroom operations, safety protocols, and process tools.
- Present work at industry conferences (e.g., ECS PRiME 2024, AVS Northern California Chapter 2024).
- Contribute to cross-functional projects and communicate results with internal teams and external partners.
- Filed multiple patents and regularly present technical findings to customers.

2020–2022 **Postdoctoral Researcher**, Helmholtz-Zentrum Dresden-Rossendorf, Germany

- Fabrication and characterization of highly stable and selective gas sensors based on black phosphorus. Funded by Federal Ministry of Education and Research (BMBF), Germany.
- Development of reconfigurable transistors based on black phosphorus and their subsequent integration into 2D circuits.

2015–2020 **Ph.D. Project**, Helmholtz-Zentrum Dresden-Rossendorf, Dresden, Germany

Supervisors Dr. Artur Erbe & Prof. Dr. Gianaurelio Cuniberti

- Fabrication and characterization of fully-encapsulated vdW (photo)transistors based on 2D flakes of InSe and GaSe.
- Investigation of electronic properties of 2D MOFs. Performed first-ever four-probe and Hall-effect measurements and reported highest mobility ($230 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) achieved in a MOF system.
- Demonstration of first-ever broadband photodetector based on a 2D MOF film.
- Set up a nanofabrication lab dedicated fully to 2D materials at the institute.

Education

2015–2020 **Ph.D., Applied Physics**

Technische Universität Dresden, Germany

Title: Charge transport in two-dimensional materials and their electronic applications.

2012–2014 **M.Sc., Advanced Functional Materials and Engineering**

Joint Degree: *University of Augsburg, Germany & Grenoble Institute of Technology, France*

GPA: 1.32 (Rank 2 in the program)

2007–2011 **B.Tech (Hons.), Metallurgical Engineering**

Indian Institute of Technology, Banaras Hindu University (IIT-BHU), India

GPA: 8.44/10 (Rank 4 in the institute)

Pre-PhD Experience

- Feb–Sep 2017 **Visiting Scholar**, Dept. of Mechanical Engineering, Columbia University, New York, USA
Supervisor Prof. Dr. James Hone
- Participated in the development of lithography-free “via-contacts”, a novel device encapsulation scheme for air-sensitive 2D materials to enhance their lifetime and performance.
 - Learnt techniques of complex vdW heterostructure stacking.
- Jan–Jul 2015 **Research Assistant**, Fraunhofer Institute for Organic Electronics, Dresden, Germany
- Responsible for the deposition and investigation of AlN and AlScN thin films using magnetron sputtering.
 - Worked closely with the industrial partners; films were produced with required piezoelectric properties, stress levels, thickness, and composition.
- 2014 **Master’s Thesis**, Group of Large Area Electronics, IMEC, Leuven, Belgium
Supervisors Dr. Pawel Malinowski & Prof. Dr. Paul Heremans
- Investigation of the origins of dark current in organic photodetectors.
 - Integrated a metal-oxide-based electron transport layer into photodetectors for the first time, and as a result, reduced the dark current and improved their performance.
- 2011–2012 **Project Manager (Operations), Cold Rolling Mill (CRM)**, Kalinganagar Project, TATA Steel, Odisha, India
- Reviewed engineering design of equipment and conducted inspections at manufacturer’s facilities.
 - Implemented safety standards and protocols on project site.
 - Selected for Central Project Monitoring team to monitor timeline and budget.
 - Collaborated with consulting firms including Accenture for project optimization.
- May–Jul 2010 **Undergraduate Intern**, R&D Dept., TATA Steel, India
- Metallurgical investigations of surface defects in cold rolled steel stripes (automobile panel grade of steel).
 - Rigorous statistical analysis to correlate defect severity and continuous casting parameters, such as mould level fluctuations, casting speed variation, and stopper head position.

Publications

h-index: 8, i10-index: 8, 979 citations ([Google Scholar link](#))

1. A. Echresh, M. S. Shaikh, M. T. Catuneanu, H. Arora, S. Zhou, A. Erbe, M. Helm, L. Rebohle, K. Jamshidi, Y. Berencén, Y. M. Georgiev, High-Performance and Ultrafast Single Germanium Nanowire Photodetectors. *Adv. Optical Mater.* **13**, 2403097 (2025).
2. H. Arora, Z. Fekri, A. Erbe. Fully Encapsulated and Stable Black Phosphorus Field-Effect Transistors. *Applied Materials Technologies* **8** (2), 2200546 (2023).
3. T. Venanzi, M. Selig, A. Pashkin, S. Winnerl, M. Katzer, H. Arora, A. Erbe, A. Patane, Z. R. Kudrynskyi, ZD Kovalyuk, L. Baldassarre, A. Knorr, M. Helm, H. Schneider. Carrier dynamics in InSe and the impact of terahertz pulses. *47th International Conference on Infrared, Millimeter and Terahertz Waves (IRMMW-THz)*, 1-2, Delft, Netherlands (2022).
4. T. Venanzi, M. Selig, A. Pashkin, S. Winnerl, M. Katzer, H. Arora, A. Erbe, A. Patane, Z. R. Kudrynskyi, ZD Kovalyuk, L. Baldassarre, A. Knorr, M. Helm, H. Schneider. Terahertz control of photoluminescence emission in few-layer InSe. *Appl. Phys. Lett.* **120**, 092104 (2022).
5. H. Arora and A. Erbe. Recent Progress in Contact, Mobility, and Encapsulation Engineering of InSe and GaSe. *InfoMat*, **3**(6), pp. 662-693 (2021).
6. A. Echresh, H. Arora, F. Fuchs, Z. Li, R. Hübner, S. Prucnal, J. Schuster, P. Zahn, M. Helm, S. Zhou, A. Erbe, Lars Rebohle, Y. M. Georgiev. Electrical Characterization of Germanium Nanowires Using a Symmetric Hall Bar Configuration: Size and Shape Dependence. *Nanomaterials* **11** (11), 2917 (2021).
7. H. Arora. Charge transport in two-dimensional materials and their electronic applications. *Ph.D. Dissertation*, Technische Universität Dresden (2020).
8. H. Arora, R. Dong, T. Venanzi, J. Zscharschuch, H. Schneider, M. Helm, X. Feng, E. Cánovas, A. Erbe. Demonstration of a Broadband Photodetector Based on a Two-Dimensional Metal-Organic Framework. *Advanced Materials* **32**, 1907063 (2020). (Featured on the [issue’s back cover](#), in *Optics and Photonics News*, [laserfocusworld.com](#) and [sciencedaily.com](#)).
9. H. Arora and A. Erbe. Recent Progress in Contact, Mobility, and Encapsulation Engineering of InSe and GaSe. *InfoMat*, **10.1002/inf2.12160** (2020).
10. H. Arora, S. Park, R. Dong, A. Erbe. 2D MOFs: A New Platform for Optics?. *Optics and Photonics News* **31**,

36–43 (2020). (Feature article in October issue).

11. T. Venanzi, H. Arora, S. Winnerl, A. Pashkin, P. Chava, A. Patanè, Z. D. Kovalyuk, Z. R. Kudrynskyi, K. Watanabe, T. Taniguchi, A. Erbe, M. Helm, H. Schneider. Photoluminescence Dynamics in Few-Layer InSe. [Physical Review Materials](#) **4**, 044001 (2020).
12. F. Kern, M. Linck, D. Wolf, N. Alem, H. Arora, S. Gemming, A. Erbe, A. Zettl, B. Büchner, A. Lubk. Autocorrected Off-Axis Holography of Two-Dimensional Materials. [Physical Review Research](#) **2**, 043360 (2020).
13. H. Arora, Y. Jung, T. Venanzi, K. Watanabe, T. Taniguchi, R. Hübner, H. Schneider, M. Helm, J. C. Hone, A. Erbe. Effective Hexagonal Boron Nitride Passivation of Few-Layered InSe and GaSe to Enhance Their Electronic and Optical Properties. [ACS Applied Materials & Interfaces](#) **11**, 43480–43487 (2019). (Featured in [Eurekalert.org](#), [Phys.org](#), and presented in IQ Innovation Prize Mitteldeutschland 2020).
14. T. Venanzi, H. Arora, A. Erbe, A. Pashkin, S. Winnerl, M. Helm, H. Schneider. Exciton Localization in MoSe₂ Monolayers Induced by Adsorbed Gas Molecules. [Applied Physics Letters](#) **114**, 172106, (2019).
15. F. Kern, M. Linck, D. Wolf, T. Niermann, H. Arora, N. Alem, A. Erbe, S. Gemming, A. Lubk. Direct Correction of Residual Symmetric Aberrations in Electron Holograms of Weak Phase Objects. [Microscopy and Microanalysis](#) **25** (Suppl 2), 98–99, (2019).
16. R. Dong, P. Han, H. Arora, M. Ballabio, M. Karakus, Z. Zhang, C. Shekhar, P. Adler, P. St. Petkov, A. Erbe, S. C. B. Mannsfeld, C. Felser, T. Heine, M. Bonn, X. Feng, E. Cánovas. High-Mobility Band-Like Charge Transport in a Semiconducting Two-Dimensional Metal-Organic Framework. [Nature Materials](#) **17**, 1027–1032, (2018).
17. H. Arora, T. Schönherr, A. Erbe. Electrical Characterization of Two-Dimensional Materials and Their Heterostructures. [IOP Conference Series: Materials Science and Engineering](#) **198**, 012002, (2017).
18. H. Arora, P. E. Malinowski, A. Chasin, D. Cheyins, S. Steudel, S. Schols, P. Hereman. Amorphous IGZO as Electron Transport Layer in Organic Photodetectors. [Applied Physics Letters](#) **106**, 143301, (2015).
19. H. Arora, A. Kumar, M. B. N. Raju, A. Dey, S. Suresh. Study of Sliver Defects on Cold Rolled Coils: Effect of Casting Process Parameters. [TATA Search](#) **2**, 209-214 (2012), ISSN-0971-5975.

Technical Skills

Expertise in fabrication and characterization of complex and hybrid device structures (transistors, photodetectors, *p-n* junctions) based on 2D materials.

Sample Preparation: exfoliation and identification of 2D flakes, deterministic assembly of vdW heterostructures inside a glovebox, O₂ plasma and UV-ozone cleaning, wire-bonding.

Nanofabrication: e-beam and photo-lithography, metal deposition techniques (sputtering, thermal and e-beam evaporation), cleanroom (class 10) experience, handling of cryogenic liquids.

Characterization: Low temperature (liq. He & N₂) and high precision electrical measurements under strong magnetic fields and focused lasers, Hall-effect and four-probe electrical measurements, Raman spectroscopy, photoluminescence, scanning electron microscopy, atomic force microscopy, optical microscopy.

Software Skills: AutoCAD, OriginLab, LaTeX, Python (basics), Gwyddion, Microsoft Office.

Professional Development: Courses on "Project planning & management", "Communication & presentation skills", "Leadership & teamwork" certified by University of Surrey and Imperial College London.

Awards / Fellowships / Honors

- Oct 2020 **Best Student Award** in the entire Ph.D. Program at the Annual Workshop of IHRS NanoNet (International Helmholtz Research School for Nanoelectronic Networks).
- Aug 2018 **1st prize in Scientific Image Competition** organized by cfaed (Center for Advancing Electronics Dresden). Won travel grant worth 500 EUR. ([Link to the announcement](#)).
- Jun 2018 **3rd prize in Science Slam on "2D or not 2D"** organized by Technische Sammlungen Dresden in collaboration with Silicon Saxony e.V.
- Nov 2016 **INSPIRE cfaed Research Grant** for research stay at Columbia University, USA.
- Oct 2016 **Best Student Paper Award** at IEEE Radio 2016 Conference out of many student entries. Presented Ph.D. work titled, "*Building electronics from two-dimensional materials*".
- Oct 2016 **Travel Grant from GFF Association at TU Dresden** for IEEE Radio 2016 Conference.
- 2015–2018 **IHRS NanoNet Fellowship** for pursuing Ph.D. degree at Helmholtz-Zentrum Dresden-Rossendorf. Stipend worth 50,000 EUR.
- 2012–2014 **Erasmus Mundus Scholarship** by the European Union for pursuing M.Sc. degree. Stipend worth 48,000 EUR with Rank 1 in the selection procedure.
- Apr 2011 **BHU Alumni Association Scholarship** for overall academic and curricula excellence during undergraduate studies at IIT-BHU.

Conferences

- Invited Talks:**
- ECS PRiME 2024 Conference, held in Honolulu (HI, USA), Oct 2024. Title: Epitaxial Growth and Development of Complimentary Field-Effect Transistors.
 - Northern California Chapter AVS Junction Technology Group, Sunnyvale (CA, USA), Dec 2024. Title: Epitaxial Growth and Development of Complimentary Field-Effect Transistors.
 - Global Summit and Expo on Graphene and 2D Materials (2DMAT), Paris (France), 2021.
 - Nanoelectronics Research Lab, UC Santa Barbara (USA), 2020.
 - Annual Workshop of IHRS NanoNet, Dresden (Germany), 2019.
- Contributed Talks:**
- 5th Graphene Workshop, Basel (Switzerland), 2021.
 - Annual Workshop of IHRS NanoNet, Friedrichshöhe (Germany), 2020.
 - International Conference of IHRS NanoNet, Dresden (Germany), 2019.
 - Falling Walls Conference, Berlin (Germany), 2018.
 - Flatlands beyond Graphene, Leipzig (Germany), 2018.
 - Gordon Research Conference on “Properties and Applications of 2D Materials and Heterostructures”, Boston (USA), 2018. (Could not attend because of delays in visa process).
 - Annual Meeting of the DPG and Spring Meeting, Berlin (Germany), 2018.
 - MRS Spring Meeting, Phoenix (USA), 2017.
 - IEEE Radio Conference, Reunion Island (France), 2016.
 - CECAM conference, “Tailor-made 2D-materials”, Bremen, Germany, 2016.
 - DPG Spring Meeting, Regensburg (Germany), 2016.
 - EFDS Workshop on Graphene, Dresden (Germany), 2015.
- Summer Schools:**
- “Frontier research in 2D materials” by Graphene & Co., Cargèse (Corsica), 2018.
 - “2D Layered Materials: synthesis, properties and applications” by EPFL Lausanne, Zermatt (Switzerland), 2016.

Teaching and Mentoring

- Apr–Aug 2016 **Laboratory supervisor, Faculty of Physics, TU Dresden**
Physikalische Grundpraktikum III, undergraduate level course
- Responsibilities included preparing and supervising the experiments, and grading students’ tests and laboratory journals.
- 2016–2018 **Mentored six graduate students**
- Training them on laboratory procedures and equipment.
 - Helping conceptualizing project ideas and experiments.
 - Training them on data analysis, evaluation of the results, and scientific writing.

Outreach and Professional Service

- Active Peer reviewer for:
 - ACS Applied Materials & Interfaces (since 09/2019)
 - Advanced Materials (since 12/2020)
 - Nano Select (since 01/21)
 - Advanced Science (since 03/2021)
 - Advanced Optical Materials (since 05/2021)
 - Small (since 05/21)
 - Journal of the American Chemical Society (since 11/2021)
- Serving on the Reviewer Board of *MDPI Crystals*.
- Member of the Centre for Advancing Electronics Dresden (cfaed) (2016–2022).
- Student representative, IHRS NanoNet fellowship program (2016–2017).

Personal Skills and Competences

- Languages: **Hindi** (Mother tongue), **English** (Proficient), **German** (Fluent).
- Member of **International Peace Slam Dresden**. <https://peaceslam.com/himani/>
 - Pitched the Peace slam initiative project at **TEDx Dresden 2018**.
 - Presented the peace slam at **Palais Sommer 2018**. The news coverage is available at: <https://www.sachsen-fernsehen.de/peace-slam-beim-palais-sommer-in-dresden/>
- Former team player of Belgium National Women’s Cricket (Feb–Aug 2014).
- Participated in mountaineering expedition led by Bachendri Pal (first Indian woman to climb Mt. Everest). Trekking to Surya Top in Himalayas (4,200 meters above sea level).